MUSINI VAMSHI KUMAR

Embedded Firmware & Linux Engineer | C / C++ / Python | Device Drivers & BSP
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PROFESSIONAL SUMMARY

Embedded Firmware & Linux Engineer with 3 years of hands-on experience developing reliable embedded software for ARM-based microcontrollers and Linux platforms. Proficient in C, C++, and Python with practical expertise in UART, SPI, I2C, and GPIO protocol implementation, FreeRTOS-based multitasking, state machine design, and hardware-software integration. Experienced in Linux user-space development, kernel-level debugging with GDB, and automated test framework development. Exposure to IEC 62304 software lifecycle processes and FMEA-based risk management in a regulated medical device environment. A collaborative engineer who thrives in cross-functional teams, committed to code quality, reliability, and continuous learning.

TECHNICAL SKILLS

Languages	C, Embedded C, C++, Python, Shell Scripting, JavaScript/TypeScript (basic), Git, SVN
Embedded & BSP	Linux user-space & kernel-level development, hardware bring-up, device driver concepts, boot processes
Protocols	UART, SPI, I2C, GPIO — implementation, debugging, and peripheral integration
MCU & Platforms	STM32 (ARM Cortex-M, HAL/LL, CubeIDE, CubeMX), Raspberry Pi, ESP32, Arduino
RTOS & Concurrency	FreeRTOS, task scheduling, ISR management, multithreading, event handling, state machine design
Debugging & Tools	GDB, strace, UART logs, oscilloscope, logic analyzer, JTAG/SWD, ST-Link, STM32CubeIDE, VS Code
Standards & Safety	IEC 62304 (software lifecycle), Software FMEA, risk management documentation, MISRA-C (learning)
Other	Azure IoT, BlueZ Bluetooth, Qt/PyQt5, React/TypeScript (basic), .NET (basic), CMSIS, Keil

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer
January 2023 – Present

iLenSys Technologies Pvt. Ltd.
Hyderabad, India

Working across embedded firmware, BSP development, regulatory documentation, and software tooling for industrial and medical-grade products delivered to global clients.

- Designed and developed embedded firmware for STM32 (ARM Cortex-M) microcontrollers using C/C++, covering peripheral drivers, sensor integration, motor control, and real-time data acquisition.
- Implemented and debugged UART, SPI, and I2C communication drivers for sensors (pH, O_2, CO_2, PM2.5, photometer) and inter-processor communication between STM32 and Raspberry Pi.
- Developed FreeRTOS-based multitasking applications with task scheduling, ISR management, and state machine design for deterministic real-time performance.
- Built Linux user-space applications on Raspberry Pi (Embedded Linux); performed kernel-level debugging using GDB, strace, and UART logs; wrote shell scripts for autostart and event-driven task execution.
- Debugged hardware-software integration issues using oscilloscopes, logic analyzers, and JTAG/SWD; improved system reliability through root cause analysis and firmware fixes.
- Designed and executed embedded software test plans; developed Python automation for UART-based test data capture and Excel report generation.
- Engaged on-site at ThermoFisher Scientific (client): led first-draft IEC 62304 risk management file uplift — performed software hazard identification, risk assessment, and mitigation documentation for regulated medical device firmware.

- Integrated BlueZ Bluetooth stack for wireless connectivity; contributed to Azure IoT cloud connectivity and remote sensor data management solutions.
- Developed a custom Microsoft Outlook add-in manifest using Yo Office generator to extend productivity tooling for internal teams.
- Currently contributing to AOP Planner — an internal web application built with React (TypeScript) and ASP.NET, gaining practical full-stack exposure alongside core embedded work.

Tech: C, C++, Embedded Linux, FreeRTOS, UART/SPI/I2C, STM32, Raspberry Pi, GDB, Python, IEC 62304, BlueZ, Azure IoT, Git, Qt/PyQt5, React/TS, .NET

KEY PROJECTS

Novasync (v1.0, v2.0, v3.0) – Unified Sensor Monitoring System *May 2024 – Feb 2025*

Compact PCB integrating pH, photometer, O_2, CO_2, and PM2.5 sensors for real-time environmental monitoring on embedded Linux. Client: Industrial IoT.

- Programmed STM32L496ZET6 to interface all sensors; established UART communication between STM32 and Raspberry Pi for reliable real-time data exchange.
- Implemented SPI and I2C protocols for sensor integration; optimized data acquisition pipelines for stable performance on embedded Linux.
- Developed PyQt5 UI on 10.1-inch display for real-time visualization; implemented camera-assisted micro-cell analysis and photometer integration.
- Wrote Linux shell scripts for application autostart and event-driven task execution; integrated BlueZ Bluetooth and Azure IoT cloud connectivity.

Chart Recorder – Electromechanical Temperature Logging System *Feb 2025 – Aug 2025*

Tamper-proof embedded Linux chart recorder for long-term temperature monitoring in refrigeration environments. Permanent paper-record backup system.

- Interfaced PT100 RTD with a 16-bit ADC via SPI for high-precision temperature measurement with calibration routines for accuracy and long-term stability.
- Programmed stepper motor and pen motor control via GPIO; implemented timers and PWM for selectable chart cycles (24h to 1 month).
- Debugged hardware-software integration using UART logs, oscilloscope, and logic analyzer; validated firmware through calibration and accuracy testing.

IEC 62304 Risk Management File Uplift — ThermoFisher Scientific (Client) *Sep 2025 – Jan 2026*

On-site regulatory engagement at ThermoFisher Scientific for IEC 62304-compliant software lifecycle documentation for medical device firmware.

- Produced first-draft risk management file: software hazard identification, risk assessment, and mitigation strategy documentation.
- Collaborated with cross-functional engineering and regulatory teams to align firmware documentation with IEC 62304 software safety classification requirements.
- Developed understanding of software lifecycle documentation, traceability, and safety-critical embedded software processes in a regulated environment.

ULT Refrigerator Data Logger & Analysis Application *Oct 2023 – May 2024*

Full-featured Python desktop application for visualising, analysing, and reporting on large sensor datasets exported from Ultra-Low Temperature (ULT) refrigeration units.

- Built end-to-end desktop app in Python using PyQt5 — interactive multi-graph UI with date-range selectors, scrollable navigation, right-click context menus, and adjustable analysis parameters.
- Processed large Excel/CSV datasets using Pandas DataFrames for fast in-memory access; implemented automatic detection and visual highlighting of unstable or anomalous sensor behaviour.
- Developed automated report generation engine — users can view or save structured reports with sensor performance insights, trends, and flagged anomalies.
- Implemented segmented data views allowing engineers to isolate specific time windows and sensor channels for root cause analysis of refrigeration faults.

ERPC LCD Display Testing Framework *May 2023*

Automated embedded test framework for LCD display verification using STM32 and UART-based ERPC communication.

- Programmed STM32 with structured test cases for UART/ERPC LCD verification; built Python test harness for automated data capture and Excel report generation.

DXF Comparison Automation Tool *Jun 2023 – Oct 2023*

Python tool automating comparison of engineering DXF drawing versions — detects and visualizes differences, reducing manual review effort significantly.

- Built DXF parsing and diff algorithm using ezdxf; rendered highlighted PNG outputs via PyQt5 GUI with robust file handling and error management.

EDUCATION

Bachelor of Technology — Electronics and Communication Engineering *2022*

ACE Engineering College, Ghatkesar | CGPA: 7.06

ACHIEVEMENTS & ADDITIONAL

- Presented a technical seminar on Embedded Systems at ACE Engineering College — focused on Raspberry Pi and real-world embedded Linux applications.
- Exposure to regulated medical device software environments (IEC 62304) and quality processes through on-site client engagement at ThermoFisher Scientific.
- Continuously upskilling in MISRA-C guidelines, KGDB kernel debugging, and networking protocols (L2/L3) to strengthen system-level embedded expertise.
- Full-stack exposure (React/TypeScript, ASP.NET) through active internal project participation — demonstrating adaptability across software domains.